

Adventure Works 2020

Power BI - Insight Report

Visual #1 - Total Revenue

Visual #1	Total Revenue (KPI Card)
What It Shows	Displays total sales revenue across all products, customers, and territories from July 2016 to July 2019 using the Total Revenue DAX measure (SUM of Sales[SalesAmount]).
Key Insight	Total revenue reached \$29.36M over three years, confirming a healthy and growing business. This is the headline figure every stakeholder looks at first.
Business Decision	Use this as the baseline benchmark for setting annual targets. Comparing it against the 98.38% Budget Attainment that company nearly met its goals, the next cycle should set a slightly more ambitious target.

Visual #2 - Total Profit

Visual #2	Total Profit (KPI Card)
What It Shows	Shows total profit calculated as Total Revenue minus Total Product Cost, reflecting how much money the business kept after covering production costs.
Key Insight	Total Profit is \$12.08M against \$29.36M revenue the business retains approximately 41 cents for every dollar earned, indicating strong cost management.
Business Decision	Monitor whether profit grows proportionally with revenue. If revenue grows but profit stays flat, it signals rising costs. The finance team should use this to assess whether pricing adjustments are needed.

Visual # 3 — Profit Margin %

Visual #3	Profit Margin % (KPI Card)
What It Shows	Displays overall Profit Margin percentage (Total Profit ÷ Total Revenue × 100), summarizing pricing and cost efficiency in a single number.
Key Insight	Adventure Works maintains a 41.15% profit margin. For every \$100 of sales, \$41.15 is earned in profit. This is exceptionally strong for a retail and manufacturing business. Business Decision
Business Decision	Protect this margin by avoiding heavy discounting. Investigate which categories and products drive the highest margins. Accessories may have strong profit margins compared to Bikes.

Visual #4 - Total Orders

Visual #4	Total Orders (KPI Card)
What It Shows	Shows total distinct sales orders using DISTINCTCOUNT on Sales[SalesOrderNumber], measuring transaction volume independently of revenue.
Key Insight	28,000 total orders were processed, giving an average order value of approximately \$1,048 consistent with a premium bicycle and equipment business.
Business Decision	Aim to increase order volume while maintaining average order value. Bundled product offers such as a bike with accessories could grow both metrics simultaneously.

Visual #5 - Monthly Revenue Trend

Visual #5	Monthly Revenue Trend (Line Chart)
What It Shows	Plots monthly revenue using Calendar[MonthName] on the X-axis and Total Revenue on the Y-axis, revealing the seasonal pattern of sales across the year.
Key Insight	Revenue peaks in June–July (~\$3.0M) and dips in January–February (~\$2.0M) consistently across all three years, confirming a predictable seasonal cycle.
Business Decision	Launch promotions in April–May to build summer momentum, and run January discount campaigns to recover the post-holiday dip. Increase stock levels by May to avoid stockouts during peak months.

Visual #6 - Revenue by Product Category

Visual #6	Revenue by Product Category (Bar Chart)
What It Shows	Compares total revenue across Bikes, Accessories, and Clothing using Products[Category] on the Y-axis and Total Revenue on the X-axis.
Key Insight	Bikes account for approximately \$28M while Accessories contribute only \$0.7M and Clothing contributes a very small share of total revenue over 95% of revenue comes from a single category.
Business Decision	This concentration is a business risk. A cross-selling programme encouraging bike buyers to add accessories is a low-cost first step toward diversifying the revenue base.

Visual #7 - Category Sales Mix

Visual #7	Category Sales Mix (<i>Donut Chart</i>)
What It Shows	Visualizes the proportional revenue share of each product category using Products[Category] in Legend and Total Revenue in Values.
Key Insight	Bikes hold 96.46% of revenue (\$28.32M), Accessories 2.39% (\$0.7M), and Clothing just over 1% the near-total dominance of Bikes is striking.
Business Decision	Set a strategic target to grow non-bike categories to at least 10% of total revenue within two years through dedicated marketing and expanded product lines.

PAGE# 02:

Visual #8 - Top 10 Products by Revenue

Visual #8	Top 10 Products by Revenue (<i>Bar Chart</i>)
What It Shows	Shows the ten highest-revenue products using a Top N filter on ProductName, with Total Revenue on the X-axis.
Key Insight	Mountain-200 Black (size 46) leads at ~\$1.37M, followed by Mountain-200 Black (42) and Mountain-200 Silver (38). The entire top 10 is dominated by Mountain-200 and Road-150 product lines.
Business Decision	These are star products ensure consistent stock availability, protect their pricing, and feature them prominently in all marketing materials.

Visual #9 - Bottom 10 Products by Revenue

Visual #9	Bottom 10 Products by Revenue (<i>Bar Chart</i>)
What It Shows	Shows the ten lowest-revenue products using a Bottom N filter, identifying commercially weak items in the catalogue.
Key Insight	The bottom 10 are all small accessories Half-Finger Gloves, Tire Tubes, Patch Kits, Racing Socks generating between \$2.4K and \$12.8K each.
Business Decision	Decide whether to bundle, promote, or discontinue these items. If they serve as entry-point purchases that lead to bigger sales, keep them. Otherwise, discontinuing simplifies the catalogue.

Visual #10 - Revenue by Subcategory

Visual #10	Revenue by Subcategory (<i>Clustered Column Chart</i>)
What It Shows	Breaks down revenue by product subcategory using Products[SubCategory] on the X-axis and Total Revenue on the Y-axis.
Key Insight	Road Bikes, confirming the business is built on performance cycling rather than casual or lifestyle products.
Business Decision	Concentrate investment in Road and Mountain Bikes while evaluating Touring Bikes and accessories for growth potential. Track subcategory trends quarterly to catch shifts in customer preference early.

Visual #11 - Profit Margin by Category

Visual #11	Profit Margin by Category (<i>Column Chart</i>)
What It Shows	Compares Profit Margin % across product categories using Products[Category] on the X-axis and the Profit Margin % measure on the Y-axis.
Key Insight	All categories show strong margins above 40%. Accessories and Clothing may carry margins comparable to or exceeding Bikes despite far lower revenue.
Business Decision	If Accessories have higher margins, growing Accessories revenue becomes doubly valuable. it diversifies the business and improves overall profitability. Run a detailed margin analysis per subcategory to confirm.

Visual #12 - Price vs Quantity Sold

Visual #12	Price vs Quantity Sold (<i>Scatter Plot</i>)
What It Shows	Maps individual products by ListPrice on the X-axis and OrderQuantity on the Y-axis, with each dot representing one product.
Key Insight	Low-priced products (<\$500) sell high volumes (4,000–5,000 units), while high-priced products sell fewer units. Some premium bikes still achieve moderate volumes, marking them as premium bestsellers.
Business Decision	Identify outliers in the top-right (high price AND high volume) these are the most commercially valuable products deserving the most investment and supply chain protection.

Visual #13 - Revenue by Product Color

Visual #13	Revenue by Product Color (<i>Stacked Bar Chart</i>)
What It Shows	Shows total revenue by product color using Products[Color] on the Y-axis and Total Revenue on the X-axis.
Key Insight	Black and Red are the dominant colors generating the highest revenues, followed by Silver and Yellow. Multi-color and White products contribute minimally.
Business Decision	Prioritize Black and Red variants in inventory purchasing. When launching new product lines, default to these colors first to maximize early sales.

PAGE # 03:

Visual #14 - Revenue by Occupation

Visual #14	Revenue by Occupation (<i>Bar Chart</i>)
What It Shows	Shows revenue split by customer occupation using Customers[Occupation] on the Y-axis and Total Revenue on the X-axis.
Key Insight	Professionals generate \$9.9M the highest of any group followed by Skilled Manual (\$6.4M) and Management (\$5.5M). Manual and Clerical workers contribute significantly less.
Business Decision	Direct premium product promotions at Management customers who have higher disposable income.

Visual #15 - Revenue by Income Bracket

Visual #15	Revenue by Income Bracket (<i>Column Chart</i>)
What It Shows	Displays revenue by customer income range using Customers[YearlyIncome] (binned) on the X-axis and Total Revenue on the Y-axis.
Key Insight	The \$50K income bracket generates the highest revenue at \$8.7M, followed by \$100K at \$4.5M. Middle-income earners (\$50K–\$100K) are the core customer base.
Business Decision	Calibrate pricing and marketing for the \$50K–\$100K segment. Mid-range bikes (\$1,500–\$3,000) will appeal most to this group. Pursue the \$100K+ segment for premium model sales.

Visual #16- Sales by Gender

Visual #16	Sales by Gender (<i>Donut Chart</i>)
What It Shows	Shows revenue split between Male and Female customers using Customers[Gender] in Legend and Total Revenue in Values.
Key Insight	The split is nearly equal Male 50.46% (\$14.81M) and Female 49.54% (\$14.55M) — confirming Adventure Works has a genuinely gender-balanced customer base.
Business Decision	Marketing and product design should continue appealing equally to both genders. Ensure color, sizing, and fit options are equally served to maintain this balance.

Visual #17 - Married vs Single Customers

Visual #17	Married vs Single Customers (<i>Donut Chart</i>)
What It Shows	Compares revenue from Married and Single customers using Customers[MaritalStatus] in Legend and Total Revenue in Values.
Key Insight	Single customers account for 51.73% (\$15.19M) versus Married at 48.27% (\$14.17M).
Business Decision	Position bikes as a family activity for Married customers and as a performance/lifestyle product for Single customers. Seasonal gift promotions can target the Married segment effectively.

Visual #18 - Revenue by Cars Owned

Visual #18	Revenue by Cars Owned (<i>Bar Chart</i>)
What It Shows	Shows revenue from customers grouped by number of cars owned using Customers[NumberCarsOwned] on the Y-axis and Total Revenue on the X-axis.
Key Insight	Customers owning 2 cars generate the most revenue, followed by those with 1 car. Customers with 0 cars are not the biggest buyers.
Business Decision	Market bikes primarily as performance and leisure products rather than commuter alternatives. Campaigns should focus on sporting achievement and adventure rather than convenience.

PAGE # 04:

Visual #19 - Revenue by Country

Visual #19	Revenue by Country (<i>Map Visual</i>)
What It Shows	Displays revenue as geographic bubbles by country with Territory[Country] in Location and Total Revenue in Bubble Size larger bubbles mean more revenue.
Key Insight	Australia is the single largest market at ~\$9.1M. The US contributes across multiple regions, while UK, France, and Germany each generate \$2.6M–\$3.4M.
Business Decision	Treat Australia as the priority market. Investigate what drives its outperformance and apply those lessons to grow European markets, which collectively represent significant untapped potential.

Visual #20 - Revenue by Region

Visual #20	Revenue by Region (<i>Clustered Column Chart</i>)
What It Shows	Breaks down revenue by sales region using Territory[Region] on the X-axis and Total Revenue on the Y-axis, providing a sub-country geographic view.
Key Insight	Australia leads at \$9.1M as a single region, US Southwest generates \$5.7M, and Northwest \$3.6M. Some US regions like Central and Northeast show near-zero sales.
Business Decision	Analyze what drives US Southwest's strong performance and replicate those strategies in underperforming regions like Central and Northeast, which represent growth opportunities.

Visual #21 - Revenue by Continental Group

Visual #21	Revenue by Continental Group (<i>Bar Chart</i>)
What It Shows	Aggregates revenue at the continental level using Territory[Group] on the Y-axis showing North America, Pacific, and Europe.
Key Insight	North America leads at \$11.4M, Pacific (Australia) at \$9.1M, and Europe at \$8.9M a relatively balanced global distribution across all three regions.
Business Decision	Europe at \$8.9M is close behind and could become the top region with investment. Expanding into additional European cycling markets like Netherlands or Scandinavia could tip the balance.

PAGE # 05:

Visual #22 - Actual vs Budget by Month

Visual #22	Actual vs Budget by Month (<i>Clustered Column Chart</i>)
What It Shows	Places actual monthly revenue and budgeted revenue side by side for each month using Calendar[MonthName] on the X-axis and both Total Revenue and Budget Total on the Y-axis.
Key Insight	The budget appears as a flat monthly target while actual revenue varies seasonally peaks in summer (~\$3M) and dips in winter (~\$2M) result in a 98.38% full-year attainment.
Business Decision	Future budgets should use seasonally adjusted monthly targets instead of flat amounts. Sales teams should not be penalized for natural seasonal lows or rewarded simply for summer peaks.

Visual #23 - Budget Attainment %

Visual #23	Budget Attainment % (<i>KPI Card</i>)
What It Shows	Displays the Budget Attainment percentage ($\text{Total Revenue} \div \text{Budget Total} \times 100$) the single most important metric on the Budget vs Actual page.
Key Insight	Adventure Works achieved 98.38% budget attainment, coming within 1.62% of its sales targets an outstanding result showing excellent alignment between planning and execution.
Business Decision	For the next planning cycle, raise the budget by 3–5% to push toward growth while keeping it achievable. Recognize the finance and sales planning teams for their forecasting accuracy.

Visual #24 - Actual vs Budget by Category

Visual #24	Actual vs Budget by Category (<i>Bar Chart</i>)
What It Shows	Compares actual revenue against budget by category using Products[Category] on the Y-axis and both Total Revenue and Budget Total on the X-axis as side-by-side bars.
Key Insight	Bikes dominate both actual and budget bars. Accessories and Clothing show small bars for both. Components shows actual revenue but no budget bar expected as budget only covers Bikes, Accessories, and Clothing.
Business Decision	Category managers for Accessories and Clothing should review whether budgets are ambitious enough. A dedicated campaign targeting \$1.5M in Accessories revenue would be a meaningful step up.

Visual #25 - Cumulative Revenue vs Budget Over Time

Visual #25	Cumulative Revenue vs Budget Over Time (<i>Line Chart</i>)
What It Shows	Shows two running total lines YTD Revenue and Budget Total plotted month by month to reveal whether the gap between actual and budget widens or narrows over time.
Key Insight	The cumulative actual line closely tracks the budget line all year, with both reaching ~\$29–30M by year end, visually confirming the 98.38% attainment with no extended periods of falling behind.
Business Decision	Monitor this chart monthly as an early warning system. If the actual line falls more than 5% below budget at any point, trigger a revenue recovery plan including promotions and pipeline reviews.